

Posters.science: Building Open Infrastructure for FAIR and AI-ready Scientific Posters

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Project Website: <https://posters.science>

Source Code: <https://github.com/fairdataihub/posters-science>

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Main Text of Abstract

Every year, scientific conferences bring together millions of researchers to showcase discoveries, exchange ideas, and build collaborations. Posters are a prevalent form of communication at these events. While no exact count is available, it is estimated that millions of scientific posters are presented every year, making them one of the most common forms of scientific communication. Beyond their sheer volume, posters play a distinctive role: they introduce preliminary insight and early findings, often for the first time, making them a valuable and timely source of scientific knowledge. This is particularly relevant in biomedical research, where conference posters often represent the earliest public glimpse of findings that will eventually shape clinical guidelines, drug targets, or genomic databases. Yet the vast majority of this knowledge evaporates when the conference ends. Walk the halls after a poster session, and you will find posters rolled under chairs, forgotten against walls, stuffed into recycling bins, or later abandoned at airport security on the way home.

Since no prior assessment of poster sharing practices existed, we conducted our own investigation. We searched online platforms where scientific posters are shared, combining Google search, Large Language Model (LLM)-assisted search, and DataCite queries. We found a total of 86 platforms where posters are shared. Across them, we could count only approximately 150,000 posters (as of December 2024), which is a tiny fraction of what we were expecting. Even when posters are shared, we found that they lack essential metadata, such as funding source or authors' ORCIDs. We also observed that many platforms do not collect conference-related information in a structured way. Even when they do, authors rarely provide it. Due to this low sharing rate and poor metadata, which makes discovery and reuse difficult, the scientific community is losing access to rich, early-stage knowledge that could inform new studies, drive collaborations, support funder decisions, and much more.

To address these challenges, we are developing Posters.science, a free, open-source platform designed to facilitate the sharing of FAIR (Findable, Accessible, Interoperable, Reusable) and AI-ready posters through intuitive user interfaces and automation. The platform reduces the process to three steps: 1) Upload the poster PDF, 2) Review and revise structured metadata automatically extracted from it by a custom LLM-based pipeline, and 3) Select a repository to publish. The platform then automatically shares the poster on the selected

repository, embedding rich metadata into the repository's record of the poster to enhance discoverability. Along with the poster's PDF, the platform also includes an auto-generated poster.json file, which is a structured, machine-friendly representation of the poster's content and metadata that ensures AI-readiness and interoperability, addressing reuse challenges that stem from the unstructured, non-standardized nature of posters. Posters.science also operates as a registry by indexing posters shared on different platforms. Each indexed poster is processed through the metadata extraction pipeline to ensure it is registered with rich metadata. By structuring poster content and metadata, the platform aims to enable novel large-scale analysis. For the biomedical research community specifically, this could open new possibilities, such as systematically mining early-stage findings on a disease or a gene across thousands of conference posters.

We initiated work on Posters.science in September 2025. As of April 2026, we have developed the schema for posters.json, based on the DataCite metadata schema with custom extensions for capturing specific metadata like conference name and location. We have also developed a first version of the LLM pipeline that automatically extracts metadata and generates posters.json from a poster PDF. Posters.science is now live, integrated with the LLM pipeline, and allows researchers to easily share posters on Zenodo, a European Union-supported generalist repository where posters are commonly shared. We have also indexed over 1,500 posters from Zenodo and Figshare, allowing users to discover them conveniently. The platform is built with Nuxt 4, Vue 3, and Nitro 2. Postgresql 18 is used as our database. We used Python 3 with Ollama to develop the LLM pipeline and metadata extraction API. The website and database are deployed via Docker containers on a Microsoft Azure virtual machine. The LLM pipeline is deployed on a local machine containing 4 RTX 3090 GPUs to provide good throughput for the end users while still maintaining reasonable costs. GitHub Actions is used to orchestrate all our CI/CD workflows. Several components of Posters.science are under active development. We are working on enabling poster sharing on the Figshare repository. We are also working to index posters from other platforms beyond Zenodo and Figshare, as well as establish periodically running poster indexing workflows. On the discovery side, we are working on an LLM-powered query tool that will allow users to ask natural language questions and get a response based on the most relevant posters from the Posters.science registry. We are also developing thorough user documentation. As usage increases, we also plan to move from a local deployment to a cloud service like Azure for the LLM model deployment.

In this talk, we will present our systematic analysis of the current poster-sharing practices, introduce the design of the poster.json schema, and give a live demonstration of Posters.science. By providing open infrastructure for sharing, indexing, and discovering posters, Posters.science aims to integrate posters into the broader open science ecosystem and make early-stage research outputs more useful to the community. We also aim to make posters more rewarding to the researchers presenting them, who are often early-stage researchers. We believe this presentation will be valuable to the BOSC community since posters represent a concrete and largely unsolved problem in open bioinformatics infrastructure. BOSC attendees produce posters every year, yet rarely have the tools to share or discover them effectively. This presentation will provide them with a simple way to change that.